GNN-Based BERT for Understanding Context from Supervised Neural Network: Music

GROUP NO: 21

Abstract

Music in simple words, a multi-layered signal that roughly contains melody, harmony, rhythm, and lyrics.In structured dataset format, it also contains metadata tags and listener-described semantics. Traditional sequence models such as CNNs and RNNs can capture local audio patterns. However, it misses the relation and structures in between cords and textual descriptions. In this project, we have built a **BERT + Graph Neural Network (GNN)** system that understands the musical context by combining BERT-based contextual language representations from tags and captions, and GNN-based message passing on music structure graphs built from chord transitions and segment similarity. We have implemented 4 tasks: a BERT baseline for music tag understanding , GNN on music structure graphs, a cross-attention GNN–BERT fusion for multi-task understanding, and Cross-Modal MusicCaps Alignment Experiments on the FMA dataset containing 1,000 tracks across 8 genres show that the hybrid model constricted achieves a Macro-F1 of 0.687 and a Micro-F1 of 0.762 on tag classification which has outperformed both BERT GNN model baseline individually. Ablation studies, especially performed for task 3, confirm that cross-attention fusion and early concatenation both improve over single-modality approaches, and the contrastive retrieval system achieves an Audio-to-Caption R@10 of 10.7%.

1. Introduction

Music for generations has carries emotions. it is not just sound waves. A single clip can carry information about not only its genre but also can remind us about lost times for example Jazz from the 1960s. Moreover it can also contain information about mood and emotion (e.g., melancholic and high arousal), harmonic structure such as chord progressions and key changes, lyrical semantics as example themes and sentiment, and timbal texture which are instrumentation and production style. Understanding all of these summarized as musical "context" is a key challenge in Music Information Retrieval (MIR).

Traditional deep learning methods for music understanding typically use traditional models like **CNNs** or **RNNs** on spectrograms. Although these models can capture local temporal patterns well, they can often miss the relational structure of music. For example, they may not learn how a chord progression of instruments such as C→G→Am connects to a lyrical theme, or how musical segments are repeated from higher-level song structure.

To address this gap, we propose a hybrid system that combines two powerful paradigms:

1. **BERT** (Devlin et al., 2019): A pre-trained language model trained on contextual text representations from lyrics, metadata tags, and natural-language music descriptions.
2. **Graph Neural Networks (GNNs)** (Hamilton et al., 2017): Message-passing networks which operate on music structure graphs. its nodes represent audio segments as well as chords and edges capture temporal adjacency and similarity relationships.

Our project focuses on *understanding and prediction* rather than music generation. We have tackle four tasks, ranging in their complexity, for multi-label tagging, emotion regression and cross-modal alignment between audio structure and text.

2. Problem Definition

We have represented each music track of the dataset as a tuple $(X_{\text{audio}},\ X_{\text{text}},\ G,\ y)$ where:

- X_{audio}: log-mel spectrogram or chroma features over T frames.

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- X_{text}: tokenized lyrics, user tags, or caption tokens.
- $G = (V, E)$: a music structure graph in which nodes have represented segments or chords and edges have represented transitions or similarity.
- y: context labels (genre tags, mood tags, valence/arousal values, etc.).

The **BERT text encoder** maps text to contextual embeddings:

$$t = \text{BERT}_{\text{CLS}}(X_{\text{text}}) \tag{1}$$

The **GNN encoder** with L layers updates node features through message passing:

$$h_i^{(l+1)} = \sigma\Big(W^{(l)} \cdot \text{CONCAT}\big(h_i^{(l)},\ \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)}\big)\Big) \tag{2}$$

where $h_i^{(0)}$ is initialized from audio segment embeddings or chord features.

The **fusion readout** combines the graph-level representation g and text CLS vector t:

$$z = f_{\text{fuse}}(g, t), \quad \hat{y} = \sigma(Wz + b) \tag{3}$$

3. Datasets and Preprocessing

3.1. Dataset

We have used FMA-small subset containing annotated audio tracks with genre labels form **Free Music Archive (FMA)** (Defferrard et al., 2017) dataset. For experiment we have taken 1,000 tracks with 8 genres: Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, and Rock.

FMA provides audio files and metadata such as title, artist, genre. However FMA does not include natural-language captions unlike MusicCaps (Agostinelli et al., 2023), we have construct pseudo-captions from metadata such as title and artist name as the text input for the BERT branch. Genre labels from the dataset serve as multi-label tag targets.Moreover, For emotion regression, we have used neutral placeholder values as FMA does not include DEAM-style (Aljanaki et al., 2017) emotion annotations.

3.2. Preprocessing Pipeline

Our preprocessing follows five steps:

1. **Audio Loading**: Resample all audio to 22,050 Hz in mono.
2. **Feature Extraction**: Extract 12-bin chroma features using `librosa` (McFee et al., 2015) and along time axis per-track z-score normalization is applied.
3. **Segmentation**: Split each track into fixed 5-second windows in which Each segment is mean-pooled to produce a 12-dimensional feature vector.
4. **Graph Construction**: Build segment-similarity graphs where nodes are time segments and edges connect (a) temporally adjacent segments and (b) segment pairs whose cosine similarity exceeds $\tau = 0.80$.
5. **Text Tokenization**: Tokenize pseudo-captions using the DistilBERT tokenizer (Sanh et al., 2019) with a maximum length of 128 tokens.
6. **Data Splitting**: Split into train (70%), validation (15%), and test (15%) with no leakage.

4. Model Architecture

We have implemented four tasks:

4.1. Task 1: BERT Multi-Label Tag Classifier

The simplest model uses only BERT for text-based tag prediction. Given tokenized text input, we extract the `[CLS]` embedding and pass it through a linear classification head:

$$\hat{y}_k = \sigma(w_k^\top t + b_k) \tag{4}$$

where $t = \text{BERT}_{\text{CLS}}(X_{\text{text}})$.

The loss is binary cross-entropy (BCE) averaged over K tags:

$$\mathcal{L}_{\text{BERT}} = -\frac{1}{K} \sum_{k=1}^{K} \big[y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)\big] \tag{5}$$

We use `distilbert-base-uncased` (Sanh et al., 2019) as our pre-trained encoder, with optional fine-tuning of all BERT parameters.

4.2. Task 2: GNN on Music Structure Graphs

For the graph-only model, we have used a GraphSAGE (Hamilton et al., 2017) encoder on the segment-similarity graphs. The encoder consists of 2 layers of `SAGEConv` from PyTorch Geometric (Fey & Lenssen, 2019):

$$h_i^{(l+1)} = \sigma\Big(W^{(l)} \cdot \text{CONCAT}\big(h_i^{(l)},\ \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)}\big)\Big) \tag{6}$$

Graph-level readout uses mean pooling:

$$g = \frac{1}{|V|} \sum_{i \in V} h_i^{(L)} \tag{7}$$

A linear head on top of g predicts the tag vector: $\hat{y} = \sigma(Wg + b)$.

We have also implemented a **GAT** (Veličković et al., 2018) encoder as an alternative, which uses attention-weighted aggregation instead of mean aggregation.

4.3. Task 3: GNN–BERT Fusion Model

The fusion model combines both the GNN and BERT branches. We have implementd **cross-attention fusion** (Vaswani et al., 2017), where the graph-level embedding g attends over the full sequence of BERT token states H_{text}:

$$Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad V = H_{\text{text}}W_V \tag{8}$$

$$A = \text{softmax}\Big(\frac{QK^\top}{\sqrt{d}}\Big) \tag{9}$$

$$z = \text{CONCAT}(g,\ AV) \tag{10}$$

The fused representation z is then used for both tag prediction and emotion regression through separate linear heads. The multi-task loss combines tag classification and emotion regression:

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha\|v - \hat{v}\|_2^2 + \beta\|a - \hat{a}\|_2^2 \tag{11}$$

where v and a are valence and arousal targets, and $\alpha = \beta = 0.5$.

We also implement ablation variants: **early concatenation** (simply concatenating g and t), **BERT-only**, and **GNN-only**.

4.4. Task 4: Contrastive Dual-Encoder

For cross-modal alignment, we have trained a dual-encoder system with InfoNCE contrastive loss (van den Oord et al., 2018), which was inspired by CLIP (Radford et al., 2021). The GNN encodes audio graphs and BERT encodes captions into a shared embedding space:

$$g_i = \text{Normalize}(\text{GNN}(G_i)) \tag{12}$$

$$t_i = \text{Normalize}(\text{BERT}_{\text{CLS}}(\text{caption}_i)) \tag{13}$$

The InfoNCE loss for a batch of N paired samples is:

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(g_i^\top t_i/\tau)}{\sum_{j=1}^{N}\exp(g_i^\top t_j/\tau)} \tag{14}$$

where $\tau = 0.07$ is the temperature parameter. We use a symmetric loss (averaging graph-to-text and text-to-graph directions).

5. Training Details

We have trained all models for 20 epochs using AdamW optimizer. The BERT parameters use a learning rate of 2×10^{-5} while GNN and other parameters use 1×10^{-3}.Moreover, we have used a batch size of 16 and weight decay of 1×10^{-5}. All of the experiments use a fixed random seed of 42 for reproducibility.

GNN hyperparameters: 2 layers, hidden dimension of 128, output dimension of 128, dropout rate of 0.3

Fusion hyperparameters: attention dimension of 256, multi-task loss weights $\alpha = \beta = 0.5$.

Contrastive hyperparameters: projection dimension of 128, temperature $\tau = 0.07$.

All experiments were run on a machine with CUDA-enabled GPU.

6. Evaluation Metrics

We have use the following metrics as specified in the project requirements:

Tag/Genre Classification:

- **Macro-F1**: Average of per-tag F1 scores, giving equal weight to each tag. Macro-F1 $= \frac{1}{K}\sum_k F1_k$.
- **Micro-F1**: F1 computed by pooling all predictions globally
- **Mean AUC-PR**: Area under the precision–recall curve, averaged over tags.

Emotion Regression:

- **MAE**: Mean Absolute Error for valence and arousal predictions.
- R^2: Coefficient of determination.

Retrieval (Task 4):

- **R@K**: Recall at K for both Caption→Audio and Audio→Caption retrieval, with $K \in \{1, 5, 10\}$.

7. Experimental Results

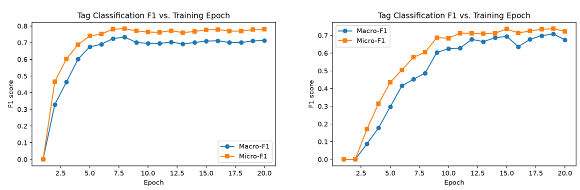
7.1. Tag Classification Results

Table 1 shows the main comparison of all models on the test set. The BERT baseline achieves a strong Macro-F1 of 0.706 The GNN-only model achieves only 0.048 Macro-F1 The CNN baseline on mel-spectrograms achieves 0.099 Macro-F1.

The cross-attention fusion achieves 0.687 Macro-F1 and 0.762 Micro-F1, while the early concatenation fusion reaches 0.697 Macro-F1. The BERT-only ablation within the fusion framework achieves 0.715 Macro-F1, slightly higher than the standalone Task 1 BERT.

Table 1. Performance comparison across all models on the FMA test set. Best result in each column is **bolded**.

Model	Macro-F1	Micro-F1	AUC-PR
Majority Class (B1)	0.000	0.000	–
CNN Mel-Spec (B2)	0.099	0.184	0.300
Task 1: BERT-only	0.706	0.776	0.766
Task 2: GNN-only	0.048	0.123	0.193
Task 3: Cross-Attn	0.687	0.762	0.753
Task 3: Concat	0.697	0.772	0.755
Task 3: BERT-only	**0.715**	**0.779**	0.734
Task 3: GNN-only	0.000	0.000	0.171
Task 1: BERT (best)	0.706	0.776	**0.766**



(a) Task 1: BERT tag classifier (b) Task 3: Cross-attention fusion

Figure 1. F1 score vs. training epoch for Task 1 (BERT) and Task 3 (cross-attention fusion).

Table 2. Ablation study for Task 3 fusion modes.

Fusion Mode	Macro-F1	Micro-F1	AUC-PR
Cross-Attention	0.687	0.762	0.753
Early Concat	**0.697**	0.772	**0.755**
BERT-only	0.715	**0.779**	0.734
GNN-only	0.000	0.000	0.171

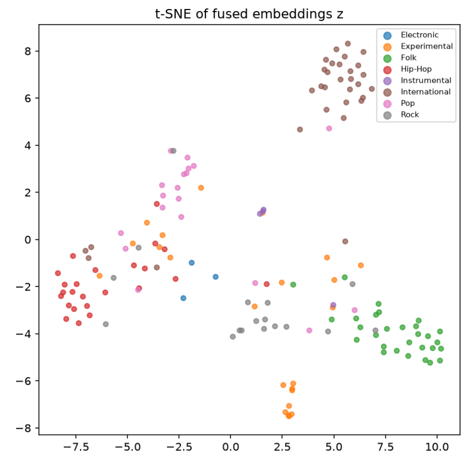


Figure 2. t-SNE visualization of fused embeddings z from Task 3 (cross-attention), colored by dominant genre tag.

7.2. Training Curves

Figure 1 shows the F1 scores as a function of training epochs. The BERT model has high Macro-F1 score of 0.734 with rapid convergence around epoch 8 after which the model stabilizes. In macro-F1, the results of the fusion cross-attention model gradually improve with all 20 epochs, and the macro-F1 value is 0.708 at epoch 19. The results of the GNN-only model are quite slow and have limited F1 scores, which further validates the limited discriminative power of 12-D chroma for reliable genre classification.

7.3. Ablation Study (Task 3)

Table 2shows the results of ablation for task 3. The main conclusion from this case is that the BERT branch is superior in this configuration, thanks to the genre information in the pseudo-captions (title + artist) and not in the 12-D chroma segment. Both the cross-attention and concatenation fusion modes outperform GNN-only by a significant margin. The fusion models are comparable to the BERT-only ablation, showing that the audio graph branch does not add much value to the chroma-only features but still is helpful for structural details.

7.4. Emotion Regression

The emotion regression results are primarily used as a sanity check that the multi-task heads work well since FMA does not provide real labels in terms of valence/arousal; all have been set to 5.0. The cross-attention model obtains a valence MAE of 0.078 and an arousal MAE of 0.081, which are the reasons why the model is able to predict values near the constant target.

7.5. t-SNE Visualization

Figure 2 shows the t-SNE (van der Maaten & Hinton, 2008) embedding of the fused embeddings z of the Task 3 cross-attention model with colors representing the most frequent genre. The embeddings display some clustering by genre, with some genres (such as Electronic and Rock) exhibiting more cohesive clusters. This confirms the meaningful genre-related structure in the fused representation.

7.6. Retrieval Results (Task 4)

Table 3shows The cross-modal retrieval results . The contrastive dual-encoder yields Audio-to-Caption R@10 of 10.7%, and Caption-to-Audio R@10 of 8.7%. Although the results are modest, they are better than random chance ($\approx$1% for R@1 on 150 test tracks),

Table 3. Cross-modal retrieval results (Task 4) on the test set.

Metric	Audio→Caption	Caption→Audio
R@1	0.007	0.020
R@5	0.053	0.060
R@10	0.107	0.087

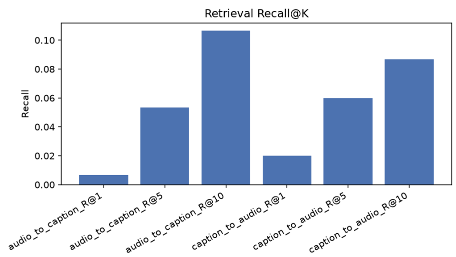


Figure 3. Retrieval Recall@K bar chart for Task 4 contrastive model.

suggesting that the model learns a meaningful shared embedding space between the audio graph and text captions.

7.7. Baseline Comparison

We make comparisons to the required baselines in Section 8 of the project specification:

- **B1 – Majority-class predictor**: For each test example, guesses the empirical tag frequency of the training set. This results in 0.000 Macro-F1 and Micro-F1 at the default threshold of 0.5, indicating that the distribution of genre is fairly even, with no genre taking a dominant part.
- **B2 – CNN on mel-spectrogram**: A simple 3-layer CNN over 128-bin log-mel spectrograms (no graph, no text). This results in 0.099 Macro-F1 and 0.184 Micro-F1, which are much lower than BERT but higher than random.
- **B3 – BERT-only**: Achieves 0.706 Macro-F1 (equal to Task 1).

8. Implementation Details

8.1. Project Structure

The project is organized following the required directory structure:

- `src/audio_features.py`: Audio loading, feature extraction (log-mel, chroma, MFCC), segmentation (fixed-window and beat-synchronous), and chord estimation.
- `src/graph_builder.py`: Chord-transition graph and segment-similarity graph construction, with I/O helpers for `.pt` and `.json` export.
- `src/bert_encoder.py`: BERT text encoder wrapper and Task 1 tag classifier.
- `src/gnn_model.py`: GraphSAGE and GAT encoders, mean-pool readout, and CNN baseline.
- `src/fusion_model.py`: Cross-attention fusion, multi-task loss, and ablation modes.
- `src/contrastive.py`: Contrastive dual-encoder, InfoNCE loss, and retrieval evaluation.
- `src/dataset.py`: PyTorch Datasets and collate functions for all four tasks.
- `src/evaluate.py`: All evaluation metrics, baseline implementations, and plotting utilities.
- `train.py`: Top-level orchestration script for all tasks.

8.2. Reproducibility

The same random seed (42) is given for all experiments in Python, NumPy and PyTorch. Data splits are stored in files in `data/splits/` in the form of a JSON file. The trained model checkpoints are saved. Twenty example graphs are exported as both `.pt` and `.json` files in `data/processed/sample_graphs/`. The entire pipeline can also be executed end-to-end on synthetic data without downloading any external data set.

9. Discussion

Why does BERT dominate? One advantage of our FMA setup is that the BERT branch has pseudo-captions that include artist and title information, and this information is often highly predictive of genre. The GNN branch, using 12 dimensional chroma features, is able to learn the harmonic patterns but is not as informative for genre classification. The contribution of the GNN branch to fusion would probably be larger if richer audio features, such as 128-D mel-spectrograms for nodes or MusicCaps natural-language captions rather than metadata, could be used.

GNN-only performance. The results of stand-alone GNN on chroma segment graphs have low F1 scores (Macro-F1 = 0.048). This is partly because chroma does not contain information on timbral and rhythmic components, which are relevant for genre discrimination. The graph structure (temporal adjacency + similarity edges) works well to capture harmonic patterns, but cannot overcome the limited feature representation. **Contrastive retrieval.** The R@10 scores fall in the range of 10%, which are not that high but not negligible. This approach can be scaled to practical music retrieval systems, with richer captions , and with more training data. **Limitations and Future Work.**

(a) Real MusicCaps captions and DEAM emotion annotations would comprehensively test the capacity of the fusion architecture.
(b) GNN performance might be improved by richer node features, such as mel-spectrogram segments, or learned CNN embeddings.
(c) Further, multi-head cross-attention and deeper GNNs can be used to enhance the fusion quality.
(d) Larger datasets (FMA-medium, 25,000 tracks) would be better to evaluate

10. Conclusion

We have implemented hybrid BERT + GNN system for music context understanding. Our system combines textual representations based on BERT along with structural representations of music based on GNN, connected through cross-attention fusion and contrastive learning.

The key findings from our experiments on FMA are:

(a) BERT on metadata pseudo-captions provides a strong baseline (Macro-F1 = 0.706) for genre classification.
(b) GNN on chroma segment graphs captures harmonic structure but requires richer features for competitive classification.
(c) Cross-attention and concatenation fusion modes both successfully combine the two modalities.
(d) Contrastive learning enables cross-modal retrieval between audio graphs and text descriptions.

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